

Calcul des dérivées du Lagrangien

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i (w \cdot x_i + b) - 1]$$

$$\frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^n \alpha_i y_i$$

$$\Rightarrow \sum_{i=1}^n \alpha_i y_i = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}(w)}{\partial w} = \nabla \mathcal{L}(w) \quad \text{Si } w = (w_1, w_2, \dots, w_d) \text{ alors}$$

$$\|w\|^2 = w \cdot w = w_1^2 + \dots + w_d^2 \text{ et on a}$$

$$\mathcal{L}(w) = \frac{1}{2} (w_1^2 + \dots + w_d^2) - \sum_{j=1}^n \alpha_j [y_j (w_1 x_j^{(1)} + \dots + w_d x_j^{(d)} + b) - 1]$$

$$\frac{\partial \mathcal{L}}{\partial w} = \nabla \mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial w_1}, \dots, \frac{\partial \mathcal{L}}{\partial w_d} \right) \text{ le gradient}$$

$$\frac{\partial \mathcal{L}}{\partial w_i} = \frac{1}{2} 2w_i - \sum_{j=1}^n \alpha_j y_j x_j^{(i)} \Rightarrow w_i = \sum_{j=1}^n \alpha_j y_j x_j^{(i)}$$

$$\frac{\partial \mathcal{L}}{\partial w} = \left(\frac{\partial \mathcal{L}}{\partial w_1} \dots \frac{\partial \mathcal{L}}{\partial w_d} \right) = 0 \Rightarrow$$

$$(w_1, \dots, w_d) = \sum_{j=1}^n \alpha_j y_j (x_j^{(1)} \dots x_j^{(d)})$$

$$w = \sum_{j=1}^n \alpha_j y_j x_j \quad (2)$$

Le problème dual

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i (w \cdot x_i + b) - 1]$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^n \alpha_i y_i} \quad (1)$$

$$\boxed{w = \sum_{j=1}^n \alpha_j y_j x_j} \quad (2) \quad \|w\| = w \cdot w$$

Par substitution de (1) et (2) dans le Lagrangien :

$$\begin{aligned} \mathcal{L}(w, b, \alpha) &= \frac{1}{2} \left(\sum_{i=1}^n \alpha_i y_i x_i \right) \cdot \left(\sum_{j=1}^n \alpha_j y_j x_j \right) - \\ &- \sum_{i=1}^n \alpha_i \left[y_i \left(\sum_{j=1}^n \alpha_j y_j x_j \right) \cdot x_i + b \right] - 1 = \\ &= \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) - \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j (x_j \cdot x_i) - \\ &- \underbrace{\sum_{i=1}^n \alpha_i y_i}_{=0 \text{ (eq. 1)}} b + \sum_{i=1}^n \alpha_i \Rightarrow \end{aligned}$$

$$\boxed{\mathcal{L} = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)}$$